

Online Appendix 2: Additional Empirical and Computational Results for Covariance-Scale Calibration of Empirical Populations

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Abstract

This Online Appendix contains additional empirical and computational analyses for the accompanying manuscript. It reports bootstrap-PCA interval widths, ridge-shrinkage sensitivity for regularized precision estimation, a gene-expression identity illustration, a split-conformal reference analysis, and detailed computational-complexity results. These materials supplement, but are not required to evaluate, the central finite-sample results and primary experiments in the main manuscript.

Keywords: covariance-scale calibration; online appendix; bootstrap PCA; ridge shrinkage; conformal prediction

1 Bootstrap PCA Interval Widths

Table 1 reports mean widths of the 95% intervals from Experiment 2 of the main paper. Because the surrogate transformation multiplies all centered observations by $\sqrt{n/(n-1)}$, it multiplies every covariance eigenvalue and percentile-bootstrap interval endpoint by $n/(n-1)$. Consequently, the surrogate percentile intervals are approximately $25/24 - 1 = 4.17\%$ wider than the corresponding ordinary percentile intervals when $n = 25$.

The surrogate increases the interval width by the deterministic scale factor without changing the relative shape of the percentile-bootstrap distribution. For λ_2 , the associated coverage increase should therefore be interpreted as a coverage-width trade-off induced by global covariance-scale correction, rather than as a separate finite-sample coverage guarantee.

Table 1: Experiment 2 mean 95% interval widths for $n = 25$, $p = 5$, $B = 2000$, and $M = 1000$. Entries are Monte Carlo means (MCSEs). The final two columns are paired width differences relative to the ordinary percentile interval based on X ; positive values indicate wider intervals.

Eigenvalue	Percentile: X	Percentile: Y	BCa: X	$Y - X$	BCa-Percentile: X
λ_1	9.68 (0.10)	10.08 (0.11)	10.33 (0.13)	0.403 (0.004)	0.653 (0.038)
λ_2	4.09 (0.04)	4.26 (0.04)	4.47 (0.04)	0.170 (0.002)	0.377 (0.016)

2 Ridge-Shrinkage Sensitivity

The precision-error comparisons in the main-paper dimension-to-sample-size analysis use the fixed ridge-shrinkage parameter $\alpha = 0.10$. To assess sensitivity to this choice, we repeat the paired precision-estimation analysis at

$$(n, p) \in \{(100, 25), (100, 100), (100, 500)\}, \quad \alpha \in \{0.05, 0.10, 0.20\}.$$

The Gaussian AR(1) design uses $\rho = 0.7$, and all configurations use $M = 1000$ paired Monte Carlo replications.

Within each replication, all values of α are evaluated using the same simulated sample. The ordinary and surrogate estimators are likewise evaluated using the same sample. Because the trace-based ridge functional is homogeneous of degree one,

$$\hat{\Sigma}_{\alpha, Y} = \frac{n}{n-1} \hat{\Sigma}_{\alpha, X}, \quad \hat{\Omega}_{\alpha, Y} = \frac{n-1}{n} \hat{\Omega}_{\alpha, X}.$$

This analysis therefore evaluates how the deterministic reciprocal scaling of the regularized precision estimate interacts with the ridge parameter and the dimension-to-sample-size ratio.

Table 2 shows that the effect of surrogate preprocessing on relative precision error is not uniform over the ridge-shrinkage parameter. At $p/n = 5$, the surrogate has lower relative precision error for all three values of α . At $p/n = 0.25$, it has lower error only at $\alpha = 0.05$ and higher error at $\alpha = 0.10$ and $\alpha = 0.20$. At $p/n = 1$, it has lower error at $\alpha = 0.05$ and $\alpha = 0.10$, but slightly higher error at $\alpha = 0.20$. Thus, the precision-estimation consequences of surrogate preprocessing depend jointly on the dimension-to-sample-size ratio and the regularization strength. These results should be interpreted as a sensitivity analysis of the specified ridge estimator, rather than as evidence for a universal precision-estimation improvement.

3 Real-Data Gene-Expression Illustration

We illustrate the surrogate transformation using the public Bioconductor ALL data package (Li, 2009). The analysis uses $n = 95$ observations designated as B-cell acute lymphoblastic leukemia in the package metadata and a deterministic panel of $p = 50$ expression probes defined as the first 50 probes in the package feature order. The feature panel is specified without using outcome labels or covariance results; the purpose of the illustration is to verify

Table 2: Robustness of relative precision-estimation error to the ridge-shrinkage parameter α . Entries are Monte Carlo means (MCSEs) over $M = 1000$ paired replications. Negative $Y - X$ values favor the surrogate under relative precision Frobenius loss. Entries shown as < 0.00005 have positive MCSEs smaller than the displayed threshold.

n	p	p/n	α	X	Y	$Y - X$
100	25	0.25	0.05	0.4113 (0.0011)	0.4063 (0.0010)	-0.0050 (< 0.00005)
100	25	0.25	0.10	0.3256 (0.0005)	0.3268 (0.0004)	0.0012 (< 0.00005)
100	25	0.25	0.20	0.4064 (0.0005)	0.4111 (0.0005)	0.0047 (< 0.00005)
100	100	1.00	0.05	1.8594 (0.0017)	1.8355 (0.0017)	-0.0239 (< 0.00005)
100	100	1.00	0.10	0.8348 (0.0009)	0.8234 (0.0009)	-0.0114 (< 0.00005)
100	100	1.00	0.20	0.4511 (0.0002)	0.4516 (0.0002)	0.0004 (< 0.00005)
100	500	5.00	0.05	4.3576 (0.0018)	4.3064 (0.0017)	-0.0513 (< 0.00005)
100	500	5.00	0.10	1.8207 (0.0009)	1.7961 (0.0008)	-0.0247 (< 0.00005)
100	500	5.00	0.20	0.6927 (0.0003)	0.6845 (0.0003)	-0.0082 (< 0.00005)

the finite-sample covariance-scale identity on an observed continuous data set, rather than to make a biological, predictive, or population-inference claim.

Let $\mathbf{S}_X$ denote the denominator- n empirical covariance of the observed expression matrix and let $\mathbf{S}_Y$ denote the corresponding covariance after surrogate preprocessing. Because the population covariance is unknown, this illustration does not estimate trace bias, population covariance loss, or confidence-interval coverage. Instead, it verifies the deterministic identities

$$\frac{\text{tr}(\mathbf{S}_Y)}{\text{tr}(\mathbf{S}_X)} = \frac{n}{n-1}, \quad \frac{\lambda_j(\mathbf{S}_Y)}{\lambda_j(\mathbf{S}_X)} = \frac{n}{n-1}, \quad j = 1, 2,$$

up to floating-point precision.

For bootstrap PCA, the ordinary and surrogate analyses use identical bootstrap resampling index vectors. Consequently, the bootstrap eigenvalue distributions and percentile-bootstrap interval widths are related by the same deterministic factor $n/(n-1)$. Table 3 verifies these relationships:

$$\frac{95}{94} = 1.010638.$$

Thus, the real-data analysis confirms that surrogate preprocessing implements the same global covariance-scale calibration outside the Gaussian simulation designs. It does not establish population-level covariance-bias correction, PCA coverage improvement, or predictive benefit for this data set.

4 Split-Conformal Reference Analysis

As a coverage-oriented reference with a distinct inferential target, we also evaluated split-conformal prediction intervals (Lei et al., 2018) using the ordinary empirical training data $\mathbf{X}$ and the same base learners as in Experiment 3 of the main paper. These results are not used for a statistical comparison with the bootstrap prediction-quantile intervals. Split conformal

Table 3: Real-data illustration of global covariance-scale calibration using $n = 95$ B-cell acute lymphoblastic leukemia samples and $p = 50$ pre-specified expression probes from the Bioconductor ALL data package. All ratios equal $n/(n - 1)$ up to floating-point precision.

Quantity	Observed ratio: Y/X	Theoretical ratio: $n/(n - 1)$
Covariance trace	1.010638	1.010638
First PCA eigenvalue	1.010638	1.010638
Second PCA eigenvalue	1.010638	1.010638
First percentile-bootstrap interval width	1.010638	1.010638
Second percentile-bootstrap interval width	1.010638	1.010638

prediction uses a training–calibration split, fits the base learner on fewer observations, and targets predictive coverage under exchangeability, whereas the bootstrap procedures quantify resampling variation in fitted predictions.

The conformal procedure uses 70% of the $n = 30$ training observations for model fitting and the remaining 30% for calibration. Within each Monte Carlo replication, the original $n = 30$ training sample is randomly partitioned once into 21 model-fitting observations and nine calibration observations; the independent test set used for evaluation remains unchanged. Thus, the conformal intervals are based on only nine calibration residuals, which produces a coarse finite-sample grid of attainable conditional coverage values.

Table 4: Split-conformal reference results for Experiment 3. Entries are Monte Carlo means (MCSEs) over $M = 500$ replications. These values are not directly compared statistically with the bootstrap prediction-quantile results, because the procedures use different training/calibration constructions and target different inferential objects.

Algorithm	Test MSE	95% interval width	95% empirical coverage
Ridge	0.48 (0.01)	2.54 (0.04)	90.3% (0.4%)
K -NN	0.95 (0.01)	3.49 (0.05)	89.3% (0.4%)

The split-conformal intervals have higher empirical coverage than the bootstrap prediction-quantile intervals in this particular small-sample design, but their observed coverage remains below the nominal 95% level: 90.3% for Ridge and 89.3% for K -NN. The mean interval widths are 2.54 and 3.49, respectively. With only nine calibration residuals per replication, finite-sample discreteness and the split-induced reduction in fitting data are substantial. These results are presented as a reference analysis with a different inferential target, not as a direct head-to-head comparison or a finite-sample coverage guarantee.

5 Detailed Computational Complexity

The surrogate transformation is computed once before the resampling or training loop. It requires calculation of the feature-wise sample mean, centering of the data matrix, scalar rescaling, and translation back to the original sample mean.

Table 5: Computational complexity of the surrogate transformation.

Step	Time	Auxiliary Memory
Validate dimensions and initialize	$\mathcal{O}(1)$	$\mathcal{O}(1)$
Compute column means	$\mathcal{O}(np)$	$\mathcal{O}(p)$
Center, scale, and translate	$\mathcal{O}(np)$	$\mathcal{O}(np)^\dagger$
Total preprocessing cost	$\mathcal{O}(np)$	$\mathcal{O}(p)$ in place

[†]If an intermediate centered matrix is materialized. An in-place implementation requires only $\mathcal{O}(p)$ auxiliary memory for the feature-mean vector.

Table 5 shows that each nontrivial operation is linear in the number of entries of the design matrix. Surrogate construction therefore requires a fixed number of linear passes over the np input entries and has total time complexity $\mathcal{O}(np)$.

If the transformation is applied in place, its auxiliary storage requirement is $\mathcal{O}(p)$ for the sample-mean vector, beyond storage of the data matrix itself. If the original observations must be retained, construction of a separate surrogate matrix requires an additional $\mathcal{O}(np)$ storage.

This preprocessing cost should be distinguished from computation performed by a downstream covariance-based or ensemble-learning procedure. Standard explicit formation of a dense $p \times p$ covariance matrix from n observations requires $\mathcal{O}(np^2)$ time and $\mathcal{O}(p^2)$ additional storage for the covariance matrix. Similarly, for an ensemble with B resamples and per-resample training cost $C_{\text{train}}(n, p)$, the surrogate-calibrated pipeline has total cost

$$\mathcal{O}(np) + \mathcal{O}(B C_{\text{train}}(n, p)).$$

Thus, surrogate construction adds a linear preprocessing cost before the resampling or training loop. For downstream procedures that explicitly form dense covariance matrices or repeatedly fit nontrivial base learners, this preprocessing does not change the leading-order computational complexity. Its practical cost should nevertheless be accounted for when data access, memory movement, or model fitting is itself close to linear in np .

References

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